Case Study Report

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Introduction

In this case study, I looked at simulations of tiny particles, such as water droplets in clouds for example, moving through turbulent air. Understanding how they form clusters can help explain things like how rain forms. The data set has 89 simulation runs with three input variables, that includes the Reynolds number (Re) which measures how turbulent the flow is, the Froude number (Fr) which tells how strong gravity is, and the Stokes number (St) which measures how much inertia the particles have. And for each run, I looked at four summary statistics, that includes the mean which is the average cluster volume, the variance which measures how much the cluster volumes differ from the average, the skewness which describes whether the distribution is more spread out toward smaller or larger clusters, and the kurtosis which tells us how likely the distribution is to contain unusually larger clusters.

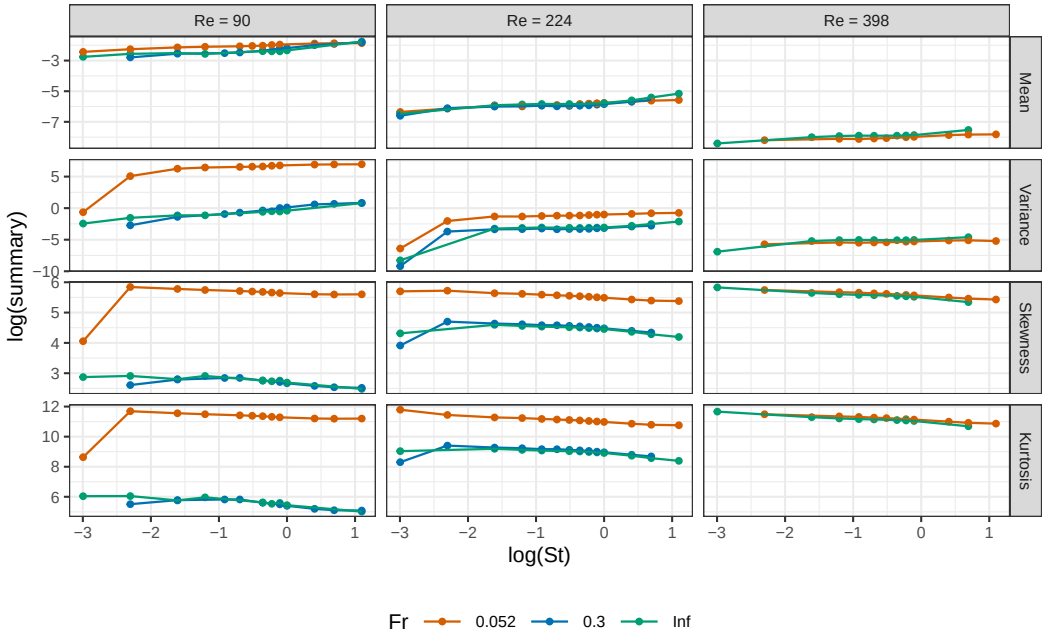
I had two objectives for this case study. The first goal is to predict these four summary statistics for new combinations of Re, Fr, and St. My second goal is to understand how each input affects the four summary statistics and whether these effects differ from one measure to another. To do this, I used log-scale liner regression models that are chosen by cross-validation.

Methodology

Preparing the data

The data contains the first four raw moments of the cluster volumes in which I used to calculate the mean, variance, skewness, and kurtosis. All four summary statistics have a huge range, such as the variance ranges from about 10^{-4} to 10^{3}. Therefore, I took the log of each statistical measure before building my models. With the data, Re and Fr only have three values each, so I treated these variables as categorical. On the other hand, St has 14 values that are mostly bunched near 0, so I used $\log(St)$ to spread out these values.

What I saw in the data



This figure showed me three important things about the data. The first observation is that the relationship with St is not linear even after taking $\log(St)$, so I added polynomial terms in my models to better capture these nonlinear patterns. The second observation is that Re and Fr interact. It is seen that strong gravity ($Fr = 0.052$) greatly increases the variance, skewness, and kurtosis at $Re = 90$, but it increases less at $Re = 224$ and it barely increases at $Re = 398$. Thus I added an Re $\times$ gravity interaction in my models to account for this relationship. The third observation is that the lines for $Fr = 0.3$ and $Fr = \infty$ are almost identical, so I replaced Fr with a simpler gravity variable that has two categories, which are strong gravity ($Fr = 0.052$) and weak or no gravity ($Fr = 0.3$ or ∞). This approach predicts just as well and is easier to interpret.

Models I tried

I started with a simple linear model and then I kept increasing its complexity. Model 1 is linear in St, Re, and gravity, and model 2 is similar but uses $\log(St)$ instead. Model 3 uses a cubic polynomial in $\log(St)$. Model 4 adds an Re $\times$ gravity interaction on top of model 3, and model 5 is similar to model 4 but it also adds an St $\times$ gravity interaction. Model 6 is the most flexible as it fits a separate St curve for each of the 6 Re-gravity combinations, while model 7 and 8 are lasso and ridge versions of model 6 that shrink its coefficients to prevent overfitting. I chose the lasso and ridge penalty with cross-validation and then fit each model separately for each summary statistic.

Picking the best model

I compared all the models with 10-fold cross-validation. I measured RMSE on the log scale, meaning that an RMSE of 0.1 meant that the predictions are usually off by roughly 10%. Since I wanted my models to be interpretable, I picked the simplest model that did nearly as well as the best one.

Final Model

The final models that selected were model 4 and model 5. The equation below is model 4. Let $x = \log(St)$. Let $R_{224} = 1$ if $Re = 224$ and $R_{398} = 1$ if $Re = 398$ (both would be 0 if Re = 90). And let $G = 1$ for strong gravity while $G = 0$ is for weak or no gravity.

$$\log(\text{summary}) = \underbrace{b_0 + b_1 x + b_2 x^2 + \cdots + b_d x^d}_{\text{curved effect of St}} + \underbrace{b_{224} R_{224} + b_{398} R_{398}}_{\text{effect of Re}} + \underbrace{b_G G}_{\text{effect of gravity}} + \underbrace{b_{224,G}\, R_{224} G + b_{398,G}\, R_{398} G}_{\text{Re}\times\text{gravity}} + \varepsilon$$

The x terms make the St curve, b_{224} and b_{398} shift this curve for $Re = 224$ or 398, b_G shifts the curve for strong gravity, and $b_{224,G}$ and $b_{398,G}$ allow the effect of gravity to be different at each Re level. For the mean, the best model is model 5, which is the same equation as above plus an St $\times$ gravity interaction. d represents how curvy the St curve can be, and is chosen by trying $d = 1$ to 6 and keeping the lowest CV error.

Results

Which model predicts the best?

	Mean	Variance	Skewness	Kurtosis
M1: linear	0.168	1.902	0.608	1.176
M2: log(St)	0.139	1.735	0.614	1.187
M3: cubic in log(St)	0.139	1.688	0.619	1.200
M4: + Re x gravity	0.101	0.791	0.229	0.410
M5: + St x gravity	0.074	0.788	0.277	0.511
M6: separate curves	0.148	0.684	0.231	0.405
M7: lasso	0.091	0.939	0.239	0.411
M8: ridge	0.377	0.971	0.465	0.893
Final model	0.074	0.778	0.219	0.391

d	Mean	Variance	Skewness	Kurtosis
1	0.098	0.972	0.235	0.408
2	0.095	0.832	0.219	0.391
3	0.074	0.791	0.229	0.410
4	0.074	0.778	0.231	0.414
5	0.075	0.786	0.231	0.415
6	0.076	0.788	0.232	0.415

The first table shows the cross-validation error for each model. I see that using $\log(St)$ and adding a curved relationship with St in model 2 and model 3 slightly improves the prediction accuracy. But I do recognized a big improvement coming from model 4, which includes the interaction between Re and gravity. This does reduce the CV error by more than half for variance (1.69 to 0.79), skewness (0.62 to 0.23), and kurtosis (1.20 to 0.41). For the mean, I see that model 5 performs even better by allowing the effect of St to change depending on gravity, which reduced the CV error from 0.101 to 0.074.

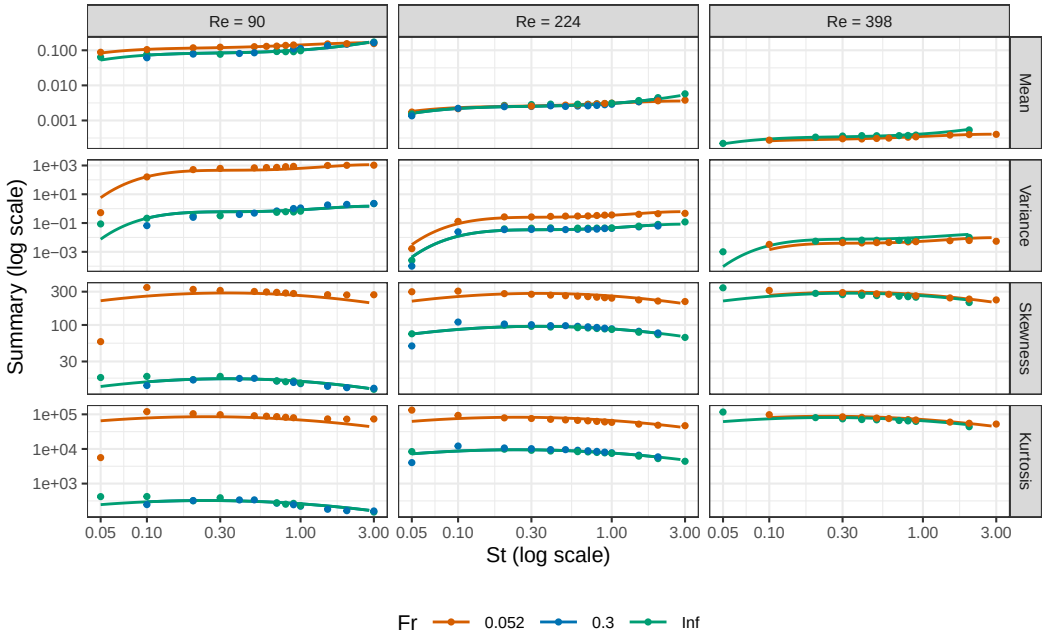
With these results, I need to consider the bias-variance tradeoff. With models 1 through 3, the models are too simple to capture important patterns in the data, which would lead to high bias. On the other hand, model 6

is too complex and would begin fitting random noise rather than the actual patterns in the data, which would result in high variance, and this could explain why the CV error for the mean increased.

In the first table, it show models 1 through 6 using a third-degree polynomial of $\log(St)$. So after selecting model 5 for the mean and model 4 for the other variables, I tested the polynomial degrees from 1 to 6 to find which degree gives the most accurate predictions as seen in the second table. The results in the second table showcase that degree 4 performs the best for the mean and variance while degree 2 performs the best for skewness and kurtosis. These choices were used to give me the final model as can be seen in the final row of the first table.

As for uncertainty, the last row of the first table gives the CV error of my final models. Since I model on the log scale, it means that our predictions are typically off by a factor of $e^{\text{CV error}}$. With our final model, the mean has a CV error of 0.074 and $e^{0.074} \approx 1.08$, meaning that my predictions are typically off by about 8%. With this in mind, the prediction errors are approximately 25% for skewness, 50% for kurtosis, and a factor of 2 for variance. But I do note that my predictions are less reliable for test points where St falls outside the range of values used to build the models for their corresponding Re and Fr settings.

How Re, gravity, and St affects the clusters



	Mean	Variance	Skewness	Kurtosis
Re 90 to 224	0.0302	0.0563	5.63	29.3
Re 224 to 398	0.135	0.221	2.98	8.53
Strong vs. no gravity, Re 90	1.43	765	16.9	264
Strong vs. no gravity, Re 224	1.04	7.42	2.96	8.68
Strong vs. no gravity, Re 398	0.866	0.525	1.03	1.09

The plots here shows my models' predictions across St, while the table showcases how much each statistical measures changes when I change one input at a time.

My first big takeaway is that Re has the biggest effect. When Re increases from 90 to 224, I see that the average becomes about 33 times smaller while the skewness increases by about 5.6 times, which suggests that stronger turbulence do lead to more smaller clusters, but there are still a few very large clusters that make the distribution less balanced. Another thing I noticed is that gravity does have a greater effect whenever the turbulence is weaker. As shown from the table, when $Re = 90$, strong gravity increases the variance by about 765 times. But when $Re = 224$, the increase is only about 7 times, and I see that gravity has almost no effect when $Re = 398$. What this could tell us is that gravity causes particles to settle and form larger clusters while stronger turbulence breaks these clusters apart. Another observation to note is that the relationship with St is not linear. And the last thing I want to point out is that the four statistical measures respond differently to the input variable. In the plot, I see that increasing Re does decrease the mean cluster volume, but it also increase skewness and kurtosis. With the graph, I can also tell that gravity has little effect in the mean but significantly affects variance, skewness, and kurtosis. What this tells me is that gravity mainly changes how much the cluster sizes differ instead of significantly affecting their average cluster volume.

Conclusion

In this case study, I used simple linear regression models on the log scale in order to predict the mean, variance, skewness, and kurtosis of particle cluster volumes based on Re, Fr, and St. I compared several models using cross-validation to see which ones made the most accurate predictions. With that, I found that model 4 performs the best for variance, skewness, and kurtosis, which includes a curved relationship with St and an interaction between Re and gravity. For the mean, I found that model 5 performs better since the model accounts for how the relationship between St and the mean cluster volume changes with the strength of gravity. I do find that these models provides accurate predictions while staying it simple enough to understand how each input variable affects the cluster volumes. However, one limitation is that some test points do have St values that are outside the range I used to train the models, which makes the predictions for those points less reliable.

My main finding is that Re has the strongest effect on cluster volumes. As turbulence increases, the average cluster volume is expected to become much smaller, but there are still a few very large clusters that makes the distribution less balanced due to the fact that the skewness and kurtosis increased. Another finding is that gravity has a noticeable effect when it is strong as its influence decreases as turbulence increases. Also, the relationship between St and all four statistical measures is nonlinear, meaning that the effects of St do not follow a linear pattern basically. And the four statistical measures respond differently to the input variables, with gravity having a greater effect on how much cluster sizes varies instead of significantly affecting their average cluster volume.